

Equivalent Expressions

Lesson 6-6

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$2x + 4$ and $2(x + 2)$: when $x = 3$, both = 10; when $x = 7$, both = 18 — always the same

Equivalent

Expressions that always have the same value.

$3x + 2x + 5$ simplifies to $5x + 5$ — fewer terms, same value

Simplify

To write an expression in a shorter, simpler way.

$3x$ and $7x$ are like terms (both x); $3x$ and $3y$ are NOT (x vs y); $3x$ and $3x^2$ are NOT (x vs x^2)

Like Terms

Terms with the same letter, like $2x$ and $5x$.

$4n + 3n = 7n$ — add the coefficients ($4 + 3 = 7$) and keep the variable (n)

Combine

To add or subtract terms with the same letter.

In $5x + 3$, the coefficient of x is 5 — when $x = 2$, the $5x$ part equals 10

Coefficient

The number in front of a letter, like the 3 in $3x$.

Key Ideas & Notes

- At the music studio, two producers wrote formulas for the cost of renting equipment.
- Producer A wrote $3x + 2x + 10$, and Producer B wrote $5x + 10$.
- They look different, but are they really?
- Let's check: if $x = 4$, Producer A gets $3(4) + 2(4) + 10 = 30$, and Producer B gets $5(4) + 10 = 30$.
- Sort each expression into the group with its equivalent expression.

Think About It

- Why did both formulas give the same answer?
- What did Producer B do differently from Producer A?
- Would they still match if $x = 10$?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Producer A wrote $3x + 2x + 10$ and Producer B wrote $5x + 10$. They look different but both give 30 when $x = 4$. Why are they equivalent?

Level 1 support

Start here: $3x + 2x$ combines to $5x$, so the two forms are equivalent.

These expressions are equivalent because $3x + 2x$ ____.

Estas expresiones son equivalentes porque $3x + 2x$ ____.

I checked by substituting $x =$ ____ and getting ____ both ways.

Lo comprobé sustituyendo $x =$ ____ y obteniendo ____ en ambas.

Word bank: Equivalent Simplify Like Terms Combine

Listen for: Students explain $3x + 2x = 5x$ (combining like terms), so both equal $5x + 10$, and verify by substituting a value like $x = 4$ to get 30 both ways.

Level 2

Is testing only $x = 4$ enough to be sure two expressions are equivalent? Justify why or why not.

- Testing one value is ____ enough because ____.
- To be sure they are equivalent, I need ____.

EXPLORE

How can you prove $2(n + 4)$ and $2n + 8$ are equivalent? What two strategies could you use?

Level 1 support

Start here: Distribute or substitute values to show the forms always match.

They are equivalent because distributing $2(n + 4)$ gives ____.

Son equivalentes porque distribuir $2(n + 4)$ da ____.

I can also check by substituting $n =$ ____ and getting ____.

También puedo comprobar sustituyendo $n =$ ____ y obteniendo ____.

Word bank: Equivalent Simplify Combine Like Terms

Listen for: A strong answer uses the distributive property ($2(n + 4) = 2n + 8$) and/or substitutes a value into both to confirm equal results.

Level 2

Are $2x + 3$ and $5x$ equivalent? Find a value of x that proves your answer and explain why one test value showing equality is not enough.

- They are ____ equivalent because at $x =$ ____ they give ____ and ____.
- One matching value is not proof because ____.

CONNECT

Client A pays $4h + 2h + 30$ and Client B pays $6h + 30$. Is the pricing fair (equivalent)? How can you prove it?

Level 1 support

Start here: Combine $4h + 2h$ to $6h$, showing both clients pay the same.

The expressions ____ equivalent because $4h + 2h =$ ____.

Las expresiones ____ equivalentes porque $4h + 2h =$ ____.

I can prove it by substituting $h =$ ____: both equal ____.

Puedo probarlo sustituyendo $h =$ ____: ambas dan ____.

Word bank: **Equivalent** **Simplify** **Combine** **Like Terms**

Listen for: Students simplify $4h + 2h + 30$ to $6h + 30$, conclude the pricing is fair/equivalent, and verify by substituting a value of h .

Level 2

A third client pays $5h + 30$. Critique whether that is the same deal, and explain for which client it is cheaper as hours grow.

- $5h + 30$ is ____ the same because ____.
- As hours grow, the ____ client pays less because the coefficient of h is ____.

Guided Examples

Example 1

Which expression is equivalent to $4x + 3x$?

Solution: $4x + 3x = 7x$. Combine the coefficients: $4 + 3 = 7$.

Answer: A. $7x$

Example 2

Which expression is equivalent to $2(m + 5)$?

Solution: $2(m + 5) = 2m + 10$ using the distributive property.

Answer: A. $2m + 10$

Example 3

Which property is shown: $3(x + 7) = 3x + 21$?

Solution: Multiplying 3 by each term inside the parentheses is the distributive property.

Answer: A. Distributive

Write About the Math

The Writing Revolution

I can explain my work using the words equivalent, simplify, like terms, and combine.

1. Kernel Sentence subject + verb

Model: Equivalent is expressions that always have the same value.

Equivalente es expresiones que siempre tienen el mismo valor.

Write a kernel sentence about equivalent. Use a subject and a verb.

Escribe una oración base sobre equivalente. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Equivalent matters in math

Equivalente importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Equivalent matters in math because ____.

Equivalente importa en matemáticas porque ____.

but
pero

Equivalent matters in math, but ____.

Equivalente importa en matemáticas, pero ____.

so
entonces

Equivalent matters in math, so ____.

Equivalente importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equivalent.
Di un hecho verdadero sobre equivalent.

Equivalent ____.

Question
Pregunta

Ask a question about equivalent.
Haz una pregunta sobre equivalent.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equivalent.
Muestra entusiasmo sobre equivalent.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equivalent.
Dile a un compañero qué hacer con equivalent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

These expressions are equivalent because $3x + 2x$ ____.

Estas expresiones son equivalentes porque $3x + 2x$ ____.

I checked by substituting $x =$ ____ and getting ____ both ways.

Lo comprobé sustituyendo $x =$ ____ y obteniendo ____ en ambas.

They are equivalent because distributing $2(n + 4)$ gives ____.

Son equivalentes porque distribuir $2(n + 4)$ da ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which property is shown: $3(x + 7) = 3x + 21$?

- A. Distributive
- B. Commutative
- C. Associative
- D. Identity

Show your work:

2. Simplify: $5n + 2n$

- A. $7n$
- B. $10n$
- C. $7n^2$
- D. $52n$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says $2x + 3$ and $5x$ are equivalent because both equal 5 when $x = 1$. Is the student correct? Explain why testing one value is not enough to prove equivalence, and find a value of x where they give different results.

Sentence starter: The student is ____ because _____. When $x = 1$: $2(1) + 3 = \underline{\hspace{1cm}}$ and $5(1) = \underline{\hspace{1cm}}$. But when $x = \underline{\hspace{1cm}}$: $2(\underline{\hspace{1cm}}) + 3 = \underline{\hspace{1cm}}$ and $5(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$. To prove equivalence, expressions must be equal for ____ values of x .

Show your work:

Reflect — Exit Ticket

Which expression is equivalent to $6n + 4 + 3n - 1$?

- A. $9n + 3$
- B. $9n + 5$
- C. $63n$
- D. $18n + 4$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Distributive — *Multiplying 3 by each term inside the parentheses is the distributive property.*
2. **Try It 2:** A. $7n - 5n + 2n = 7n$. *Add the coefficients: $5 + 2 = 7$. The variable stays as n .*
3. **Exit Ticket:** A. $9n + 3 - 6n + 3n = 9n$ and $4 - 1 = 3$, *so the simplified expression is $9n + 3$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equivalent is expressions that always have the same value.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).